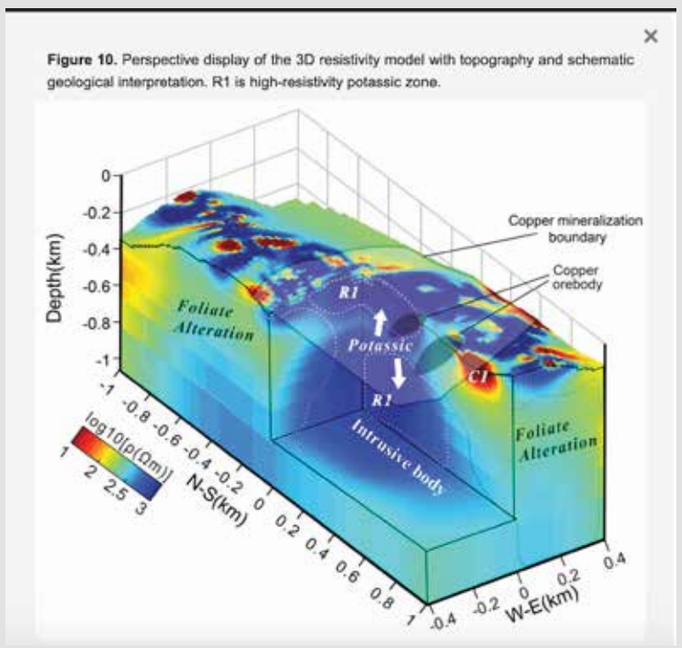
MT resistivity



https://www.mdpi.com/2075-163X/11/12/1424

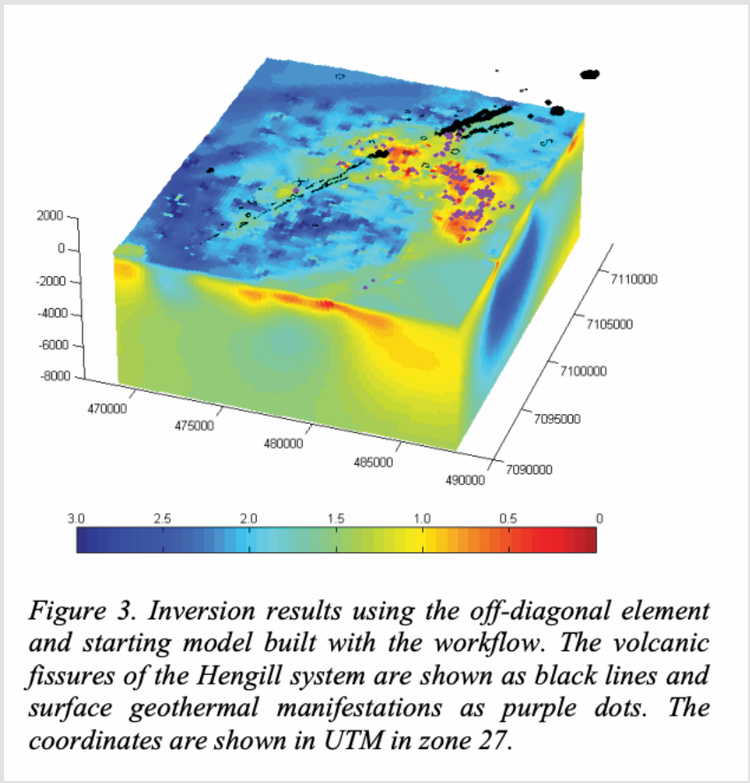


Figure 3. Inversion results using the off-diagonal element and starting model built with the workflow. The volcanic fissures of the Hengill system are shown as black lines and surface geothermal manifestations as purple dots. The coordinates are shown in UTM in zone 27.

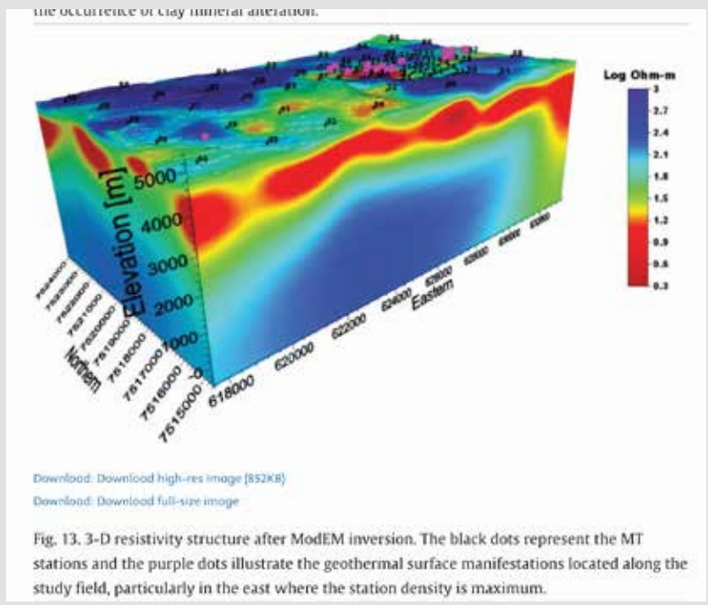
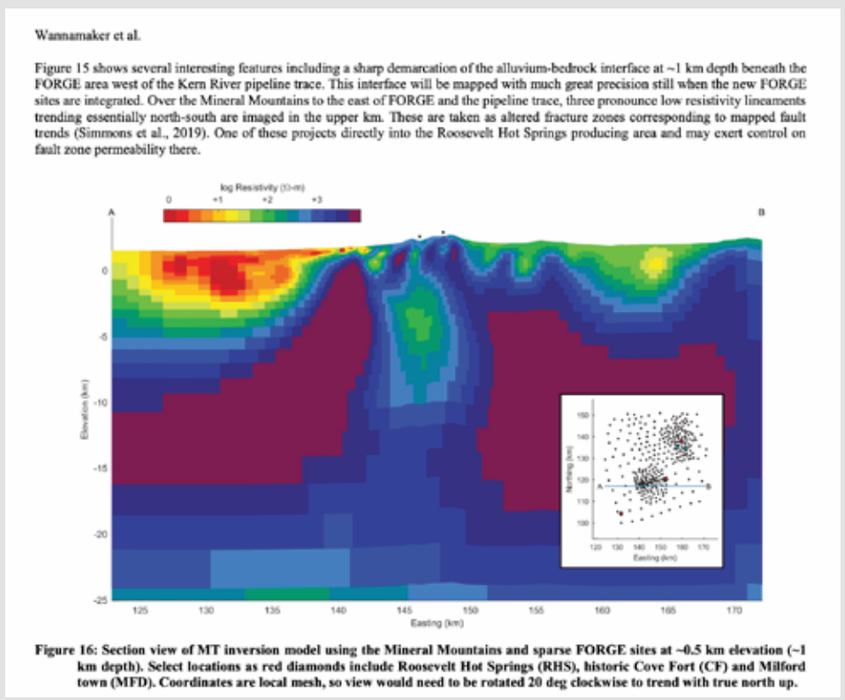


Fig. 13. 3-D resistivity structure after ModEM inversion. The black dots represent the MT stations and the purple dots illustrate the geothermal surface manifestations located along the study field, particularly in the east where the station density is maximum.

https://www.sciencedirect.com/science/article/pii/S0375650523001025

https://pangea.stanford.edu/ERE/pdf/IGAstandard/SGW/2012/Rosenkjaer.pdf

Utah FORGE Site magnetotelluric resistivity



Wannamaker et al.

Figure 15 shows several interesting features including a sharp demarcation of the alluvium-bedrock interface at ~1 km depth beneath the FORGE area west of the Kern River pipeline trace. This interface will be mapped with much great precision still when the new FORGE sites are integrated. Over the Mineral Mountains to the east of FORGE and the pipeline trace, three pronounce low resistivity lineaments trending essentially north-south are imaged in the upper km. These are taken as altered fracture zones corresponding to mapped fault trends (Simmons et al., 2019). One of these projects directly into the Roosevelt Hot Springs producing area and may exert control on fault zone permeability there.

Figure 16: Section view of MT inversion model using the Mineral Mountains and sparse FORGE sites at ~0.5 km elevation (~1 km depth). Select locations as red diamonds include Roosevelt Hot Springs (RHS), historic Cove Fort (CF) and Milford town (MFD). Coordinates are local mesh, so view would need to be rotated 20 deg clockwise to trend with true north up.

This cross-section is what a single slice through the 3D resistivity volume looks like. The Geovox demo goal would be to make this kind of data navigable and interactive — letting a client fly through it rather than looking at a static 2D figure. That contrast is a powerful selling point.

https://pangea.stanford.edu/ERE/pdf/IGAstandard/SGW/2020/Wannamaker.pdf

This figure is from a different method (Gravity meter) but it shows how a volumetirc cube would interact with topology.

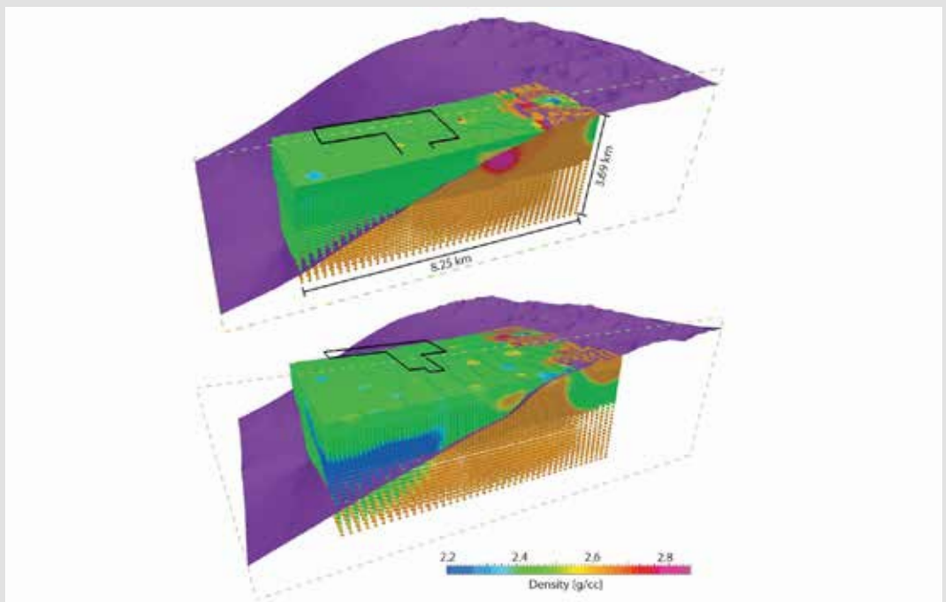


Figure 6: East-west density cross sections from 3D regularized inversion of gravity data over the FORGE site. Modified from Hardwick and Hurlbut (2019) after Witter (2019).

A good fit to the gravity data assuming a tight error floor of 0.03 mgal (Witter, 2019) and example cross sections from the inversion study appear in Figure 6. Gravity modeling experiences strong non-uniqueness and so the basement interface constraint is key. In the upper Figure 6 section close to the FORGE site, adherence to a steadily dipping interface yields a localized high-density feature in the shallowest basement correlated with local diorite rich volumes in the granitic basement. There is a slight "leakage" of the high density

5

This figure is a compelling reference for Geovox because it illustrates almost exactly the kind of scene we're aiming to deliver to clients. It shows a 3D gravity density inversion model over the FORGE site, rendered as a volumetric block with a draped topographic surface and two perpendicular cross-sections cutting through the interior — colour-scaled to make subsurface density variations immediately readable. The same approach applies directly to MT resistivity or seismic amplitude data. What's particularly useful here is that this comes from a published academic paper on the Utah FORGE site, which means the data and the visualisation approach are already validated. Geovox's pitch to a client like DMT becomes straightforward: this is what researchers produce as a static figure on a page — Geovox makes it navigable, interactive, and spinnable in real time. That contrast is one of the clearest and most immediate ways to communicate what Geovox actually does differently.